

NEREUS PROTOCOL – EIC Accelerator Strategic Review

This document contains:

1. Strategic evaluation of the existing NEREUS EIC Accelerator deck
2. Key weaknesses and risks from an evaluator perspective
3. Suggested positioning improvements
4. A revised EIC-oriented narrative structure

Overall Assessment

The concept is strategically strong and operationally credible, particularly because it originates from real maritime and disaster-response experience rather than purely academic or theoretical assumptions. The strongest aspect of the project is the infrastructure-collapse communication thesis: existing systems fail precisely when disasters become most critical. The project aligns well with: - EU resilience strategy - Civil protection modernization - Infrastructure sovereignty - Decentralized emergency coordination - Low-cost interoperability

Primary Weaknesses

The current deck presents the project as more operationally validated than it actually is. This creates unnecessary evaluator risk. Key missing elements: - Large-scale field validation - Real packet-loss metrics - Maritime RF stress testing - Certification roadmap detail - Security/authentication architecture - Institutional pilot partners - Technical benchmark data The recommendation is not to exaggerate validation maturity. Instead, the proposal should frame EIC funding as the mechanism enabling real-world validation.

Regulatory Concerns

The proposal touches heavily regulated areas: - VHF Channel 70 usage - DSC compatibility - MMSI assignment logic - ETSI compliance - Maritime communication standards Particularly, automatic MMSI-style assignment may create concerns among evaluators familiar with maritime regulation. A safer framing would emphasize: - temporary disaster-mode identifiers - compatibility with existing frameworks - emergency-only operation modes - future regulator collaboration

Strategic Recommendation

The proposal should move away from a “world-changing protocol” tone and toward a “serious European resilience infrastructure” tone. Recommended positioning: - technically honest - validation-aware - phased deployment - engineering-driven - operationally grounded - regulation-conscious This improves evaluator trust significantly.

Alternative EIC-Oriented Narrative Structure

Core Thesis

Modern disaster response systems fail when infrastructure fails. NEREUS creates a decentralized emergency coordination layer operating entirely over existing VHF radio infrastructure — enabling vessels, volunteer units and SAR operators to self-organize without internet, GSM or satellite dependency.

The Problem

During major disasters: - GSM infrastructure collapses - AIS stations become unavailable - Internet connectivity becomes unreliable - Civilian first responders remain invisible - Multi-agency interoperability breaks down Critical operational capacity exists physically — but remains invisible operationally.

The Solution

NEREUS transforms standard VHF radios into self-reporting emergency coordination nodes. The system enables: - offline unit discovery - decentralized clustering - role broadcasting - low-bandwidth tactical coordination - infrastructure-independent situational awareness

Current Technical Status

Estimated maturity: TRL 4–5 Validated components: - Bell 202 AFSK transmission architecture - packet structure logic - decentralized clustering concepts - offline coordination workflows Pending validation: - large-scale maritime testing - certification - interoperability validation - field deployment

Why Now

Europe is entering a strategic resilience phase. The project aligns strongly with: - Critical Infrastructure Resilience Directive - EU Civil Protection modernization - Infrastructure sovereignty - Disaster resilience initiatives - Low-cost decentralized communication systems

Strategic Vision

NEREUS is not intended to become another proprietary communication device. The long-term objective is to establish: an open, resilient emergency coordination protocol layer for infrastructure-collapse scenarios. “No Signal” should no longer mean “No Coordination.”